Supplementary Material
Bayesian VAR with lag selection

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Scope. This document accompanies the main article and provides the numerical specification of its retained three-break simulation, the associated collapsed-MCMC initialization and diagnostics, and supporting diagnostics for the quarterly U.S. macroeconomic application. It also preserves the one-break, two-break, and simulated break-count-selection experiments from the thesis that were excluded from the main article, including their original tables and figures. These additional results are documented separately in Sections S7–S9, and should not be interpreted as additional runs carried out for this supplement. Table and figure values originate in the thesis from which the article is derived and are not presented as results of a new numerical execution. The source R code and original input snapshots are preserved in the separate paper repository specified in Section S10.

S1 Three-Break Simulation: Data-Generating Process

S1.1 Sample, regimes, and lag configurations

The retained experiment is a bivariate piecewise VAR with maximum lag order $K = 3$ and exactly three break dates. After discarding 300 initial simulated observations, $T = 3{,}400$ observations are retained: $T_0 = 200$ are used as a fixed training sample, followed by $T_{\text{est}} = 3{,}200$ estimation observations. The true break positions within the estimation sample are $(800, 1600, 2400)$, corresponding to full retained-sample positions $(1000, 1800, 2600)$. Consequently, each estimation regime has 800 observations. The true blockwise lag configurations are 100, 010, 001, and 110, in regime order.

For $s = 1, \ldots, 4$, the generated observations satisfy

$$y_t = c_s + \sum_{k=1}^{3} B_{k,s} y_{t-k} + \varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}_2(0, \Sigma_s), \tag{1}$$

where s is the regime containing observation t. The active coefficient matrices are

$$A_1 = \begin{pmatrix} 0.45 & 0.10 \\ -0.05 & 0.35 \end{pmatrix}, \quad A_2 = \begin{pmatrix} -0.35 & 0.15 \\ 0.10 & 0.25 \end{pmatrix}, \quad A_3 = \begin{pmatrix} 0.18 & 0.00 \\ 0.08 & -0.15 \end{pmatrix}. \tag{2}$$

The three matrices used in each regime, ordered by lag $k = 1, 2, 3$, are

$$(B_{1,1}, B_{2,1}, B_{3,1}) = (A_1, 0, 0), \tag{3}$$
$$(B_{1,2}, B_{2,2}, B_{3,2}) = (0, A_2, 0), \tag{4}$$
$$(B_{1,3}, B_{2,3}, B_{3,3}) = (0, 0, A_3), \tag{5}$$
$$(B_{1,4}, B_{2,4}, B_{3,4}) = (A_1, A_2, 0), \tag{6}$$

where every zero denotes a 2×2 zero matrix. The regime-specific intercepts are

$$c_1 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \quad c_2 = \begin{pmatrix} 0.20 \\ -0.10 \end{pmatrix}, \quad c_3 = \begin{pmatrix} -0.15 \\ 0.15 \end{pmatrix}, \quad c_4 = \begin{pmatrix} 0.10 \\ 0.10 \end{pmatrix}. \tag{7}$$

The four innovation covariance matrices are

$$\Sigma_1 = \begin{pmatrix} 1.00 & 0.30 \\ 0.30 & 1.00 \end{pmatrix}, \qquad \Sigma_2 = \begin{pmatrix} 0.70 & -0.20 \\ -0.20 & 1.20 \end{pmatrix}, \tag{8}$$

$$\Sigma_3 = \begin{pmatrix} 1.10 & 0.15 \\ 0.15 & 0.80 \end{pmatrix}, \qquad \Sigma_4 = \begin{pmatrix} 0.90 & 0.00 \\ 0.00 & 0.90 \end{pmatrix}. \tag{9}$$

S1.2 Stability and prior calibration

The spectral radii of the four regime-specific companion matrices, computed from the full three-lag representation (with zero matrices for excluded lags), are 0.4031, 0.6116, 0.5646, and 0.7284, respectively. Thus each individual regime is stable under its stated parameter values. These are properties of the simulated DGP, not stability claims for arbitrary posterior coefficient draws.

The DGP is generated with R random seed 2026. Inference fixes the number of breaks at $m = 3$, imposes a minimum admissible regime length of 200 observations, and uses the first 200 retained observations for the common prior calibration. The numerical prior choices in the definitive R implementation are $\lambda_{\text{const}} = 10$, $\lambda_{\text{lag}} = 0.40$, $\lambda_{\text{decay}} = 1$, own-lag prior mean $\mu_{\text{own}} = 0$, and inverse-Wishart degrees of freedom $\nu_0 = n + 2 = 4$ for $n = 2$. The shared prior calibration is fixed rather than re-estimated separately for each break proposal. See the main article for the complete Matrix-Normal–Inverse-Wishart formulation and model-marginalized segment evidence.
Provenance: thesis Section 5.1.3, pp. 77–80; original R source, three-break setup beginning at the block marked "29. Execution: a new $m = 3$ case."

S1.3 Numerical safeguards for prior calibration

The fixed training-sample prior calibration described in Section 2.3 of the main article uses a preliminary VAR(1) regression with an intercept. In the definitive R implementation, a ridge term $10^{-8}I$ is added to the regressor cross-product matrix when estimating its coefficients. The divisor for the residual covariance estimate is $\max\{T_0 - 1 - (n + 1), 1\}$: the lower bound of one is a numerical safeguard and is inactive in the samples reported here. The training covariance estimate and inverse-Wishart prior scale are symmetrized and, if needed, diagonally adjusted to ensure positive definiteness. These operations are numerical safeguards for the prior construction, not changes in the regime-specific Bayesian VAR model. They are implemented by `make_prior_info()` and `make_pd()` in `code/paper/common.R` of the current paper repository. No additional estimation was performed in preparing this description.

S2 Final Dispersed-Start MCMC Specification

S2.1 Chain initialization and sampling schedule

The final simulation uses the collapsed MCMC sampler with $m = 3$ fixed, four dispersed-start chains, 150,000 iterations per chain, and 25,000 burn-in iterations per chain. The remaining 125,000 observations from each chain are retained without thinning for posterior summaries and diagnostics. Thinning by a factor of five is used for trace plotting and reduced graphical representations only. The thesis expresses initial break locations in estimation-sample indices, whereas the archived R implementation stores break locations in full retained-sample indices. Because the common training sample has length $T_0 = 200$, these are the same starting configurations expressed on different scales: $\tau_{\text{full}} = T_0 + \tau_{\text{est}}$. Table S1 reports both coordinate systems for transparency. The initial lag configurations, iteration counts, and burn-in settings coincide; the seeds are explicitly recorded in the R code. The 300-observation DGP burn-in and the 25,000-iteration MCMC burn-in are distinct from this indexing offset.

Table S1: Identical dispersed-chain starting configurations expressed in estimation-sample (thesis) and full retained-sample (R code) indices.

Chain	Seed	Breaks: estimation scale	Breaks: full scale	Initial lag models (regimes 1–4)
1	1001	$(450, 1300, 2350)$	$(650, 1500, 2550)$	111, 111, 111, 111
2	1002	$(650, 1500, 2500)$	$(850, 1700, 2700)$	000, 000, 000, 000
3	1003	$(300, 1100, 2100)$	$(500, 1300, 2300)$	101, 010, 110, 001
4	1004	$(800, 1700, 2700)$	$(1000, 1900, 2900)$	011, 101, 001, 111

Sources: thesis Section 5.1.3 (thesis starting positions and initial lag models); `BVAR_thesis_full_code.R`, dispersed-start block, `breaks_init_list_3b`, `gammas_init_list_3b`, and `seeds_disp_3b` (implemented starts and seeds). The thesis prints break locations relative to the estimation sample; the R sampler internally uses the indices of the full retained sample, including $T_0 = 200$ training observations. Subtracting T_0 from each R starting value yields exactly the thesis entry in the same row. The true break positions are $(800, 1600, 2400)$ on the estimation scale and $(1000, 1800, 2600)$ on the full scale. This coordinate conversion resolves the apparent starting-state discrepancy; independent numerical reproduction is still pending.

The MCMC produces a posterior over break locations and lag configurations *conditional on three breaks*; it is not an exact search over alternative break counts. The empirical application of the article instead uses exhaustive evaluation for $m \leq 3$ and exact dynamic programming for $m = 4$ to maximize the model-marginalized evidence over admissible break partitions. These inferential procedures are kept separate. The numerical acceptance probabilities and move mechanisms are those of the original implementation, with no newly introduced proposal method.

S2.2 Multi-chain break traces and convergence

Figure S1 reproduces the three multi-chain trace plots published as thesis Figure 5.6. The starting states in Table S1 coincide after adjusting for the $T_0 = 200$ training-sample offset; reproducing the plotted draws by a fresh run of the archived R script has nevertheless not been independently verified. The dashed horizontal reference lines represent the true break positions. The four chains overlap considerably for the first and third breaks. The second break occasionally moves to earlier locations and has the weakest sampling efficiency: its bulk-ESS is 882.07 and its tail-ESS is 395.82, despite rank-normalized $\widehat{R} = 1.0050$. These are the diagnostics from the unthinned post-burn-in chains; the plotted trajectories alone are thinned. The visible overlap does not prove complete posterior exploration.

S3 Additional Three-Break Posterior Diagnostics

S3.1 Individual lag indicators

Table S2 preserves the full indicator-by-indicator diagnostics reported in thesis Table 5.9. These are published numerical summaries, not newly recomputed diagnostics from the archived code. Each indicator pertains to the inclusion of a *whole lag-coefficient matrix* in one regime, not an equation-specific coefficient. The minimum transition count is the minimum over the four chains; an entry of zero may therefore coexist with some switching in other chains. The thesis reports undefined $\widehat{R}$ or ESS in some near-constant indicator components, shown as "NA" rather than replaced with invented values.

The truly active lags have PIPs reported as 1.000. Among inactive lags, inclusion probabilities are largest for lag 1 of regime 3 (0.088) and lag 3 of regime 2 (0.066). For these components, the minimum transition counts are 879 and 662, respectively. This supports actual switching between the two indicator states, although no such inference follows automatically for indicators with zero transitions or undefined efficiency diagnostics.

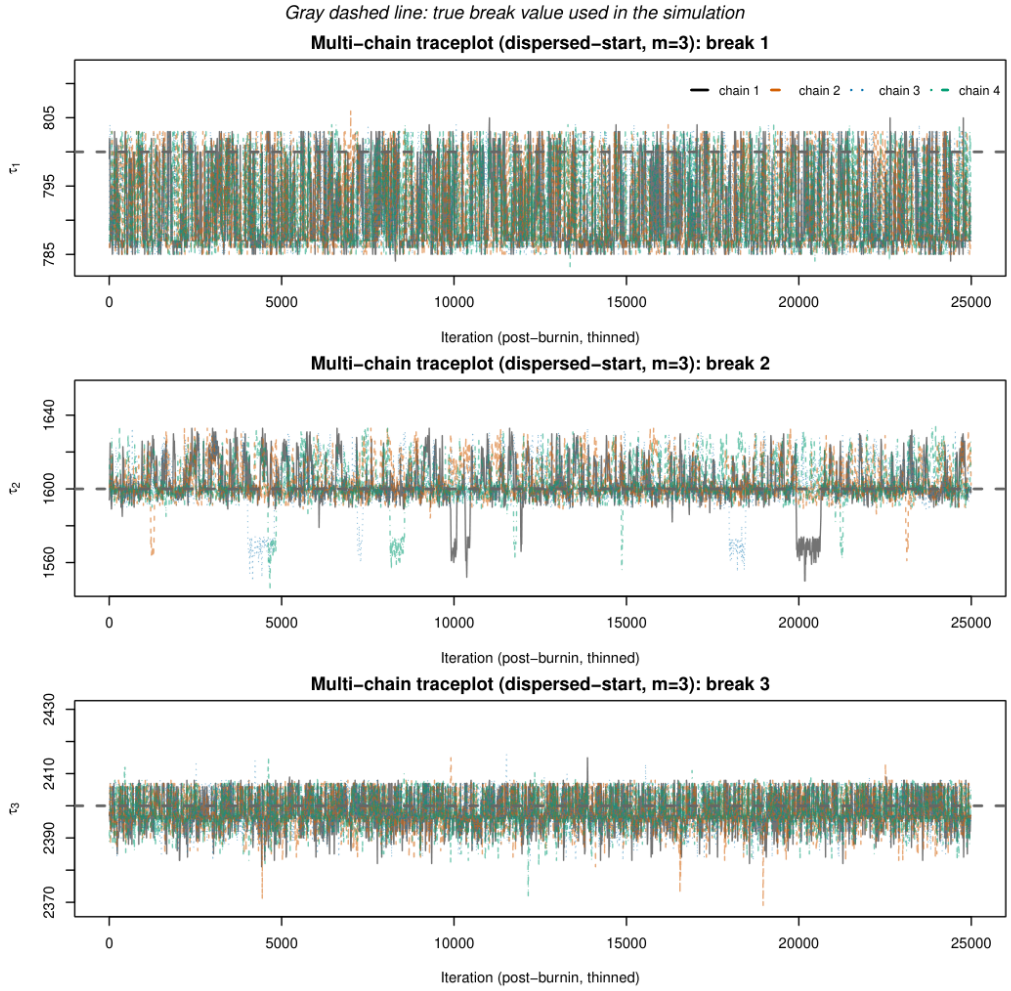


Figure S1: Multi-chain post-burn-in traces for the three simulated break positions. Extracted from thesis Figure 5.6 (same plotted trajectories, not newly generated). Horizontal dashed lines indicate true break positions. The horizontal axis displays thinned post-burn-in iterations.

Table S2: Posterior inclusion probabilities and multi-chain MCMC diagnostics for all 12 lag indicators.

Regime	Lag	PIP	$\widehat{R}$	Bulk-ESS	Tail-ESS	$T_{\min}$
1	1	1.000	NA	NA	NA	0
1	2	0.002	1.001	8171.9	8171.9	8
1	3	0.020	1.000	10846.4	10846.4	208
2	1	0.000	1.000	26001.4	26001.4	0
2	2	1.000	NA	NA	NA	0
2	3	0.066	1.000	11662.1	NA	662
3	1	0.088	1.000	11589.0	NA	879
3	2	0.017	1.001	10293.0	10293.0	153
3	3	1.000	NA	NA	NA	0
4	1	1.000	NA	NA	NA	0
4	2	1.000	NA	NA	NA	0
4	3	0.011	1.001	10574.8	10574.8	78

Notes: Source is thesis Table 5.9. Entries retain the numerical precision and NA conventions of that table. $T_{\min}$ is the minimum number of binary inclusion/exclusion transitions observed among the four chains. Values rounded to 0.000 or 1.000 are not proof that the true posterior inclusion probability is exactly zero or one.

S3.2 Proposal acceptance rates

Table S3: Acceptance rates of the four final dispersed-start MCMC chains.

Chain	Total	Break-date proposals	Lag-configuration proposals
1	0.25478	0.47536	0.03420
2	0.25486	0.47571	0.03401
3	0.25696	0.47992	0.03400
4	0.25674	0.47997	0.03351
Average	0.25584	0.47774	0.03393

Source: thesis Table 5.10. Rates use the full chain runs, including burn-in, as defined in the original implementation. Acceptance of lag-configuration proposals is much lower than acceptance of break-date proposals; these rates alone do not establish or refute adequate mixing.

The reported break and lag-model summaries are retained in the main article (its Tables 4.1 and 4.2), and are not duplicated here. There is no exact three-break posterior benchmark in the thesis. No frequentist coverage or repeated-sampling recovery probability is inferred from this single DGP realization.

S4 Data Preparation and Empirical Sample Accounting

The empirical application uses the original FRED data snapshots for the PCE price index (`PCECTPI`), the U.S. unemployment rate (`UNRATE`), and the federal funds rate (`FEDFUNDS`), all transformed to quarterly frequency. PCE inflation is defined as $400(P_t/P_{t-1}-1)$, a simple quarterly percentage change annualized by multiplication by four. Unemployment and the funds rate are retained in their original percentage units after averaging available monthly readings within a calendar quarter; at least two valid monthly observations are required for a quarterly mean. In the archived `UNRATE` snapshot, October 2025 is missing while November and December are present, so the reported 2025Q4 average uses those two observations. No alternative downloaded data vintage was substituted in constructing this supplement.

The quarterly sample spans 1954Q3–2026Q2 (288 observations), with 1954Q3–1964Q2 (40 quarters) fixed for prior calibration and 1964Q3–2026Q2 (248 quarters) for the estimation calendar. Candidate break counts are $m \in \{0, 1, 2, 3, 4\}$ and a minimum of 28 *calendar quarters* per regime is imposed. The observed 2020Q2 vector remains in the calendar and raw descriptive plots but is excluded from the likelihood. Responses using that quarter as a regressor are likewise excluded using a common valid-row mask across candidate lag configurations; observations are not imputed.

For $K = 4$, the omitted effective-likelihood responses span 2020Q2–2021Q2 (five quarters) and 243 of the original 248 estimation responses remain. For $K = 8$, the affected range is 2020Q2–2022Q2 (nine quarters) and 239 estimation responses remain. These are differences in *effective likelihood samples*, not differences in the calendar break partition.

For each candidate number of breaks, the reported empirical score is the *maximum* of the segment-integrated, lag-model-marginalized evidence over admissible break partitions. It is not an integral over break dates or a posterior probability for the break count. The selected empirical break dates are 1985Q1, 2008Q3, and 2015Q4, under both $K = 4$ and $K = 8$. A date is the *last quarter of the preceding regime*. Lag-model posterior probabilities and inclusion probabilities are calculated conditional on this selected partition.

Table S4: Selected empirical regime lengths and effective likelihood observations.

Regime	Calendar span	Calendar quarters	Effective $K = 4$	Effective $K = 8$
1	1964Q3–1985Q1	83	83	83
2	1985Q2–2008Q3	94	94	94
3	2008Q4–2015Q4	29	29	29
4	2016Q1–2026Q2	42	37	33
Total	1964Q3–2026Q2	248	243	239

Sources: thesis Table 6.2 for calendar spans and the $K = 4$ effective sample; thesis Section 6.4 and original R implementation for $K = 8$ exclusions and the last-regime effective count. Both K specifications select the same calendar break dates, but fit the likelihood over different row sets.

S5 Further Empirical Residual Diagnostics, $K = 4$

S5.1 System-level serial-correlation tests

After selecting the modal restricted VAR in each regime, the thesis refits that restricted model by OLS for the multivariate LM test. Thus the residuals for Table S5 are *not* the posterior-mean BVAR residuals used in the main article's equation-wise Shapiro–Wilk, Ljung–Box and ARCH–LM tests.

Table S5: Multivariate residual serial-correlation LM test p-values, baseline $K = 4$.

Regime	Auxiliary lag $h = 1$	$h = 2$	$h = 3$	$h = 4$
1	0.0792	0.0019	< 0.0001	< 0.0001
2	0.4087	0.1482	0.0721	0.0136
3	0.0205	0.0006	< 0.0001	n.a.
4	0.3255	0.0651	0.1166	0.0054

Source: thesis Table 6.7. Tests hold the selected break dates and lag restrictions fixed and use a restricted-OLS refit. The regime-3 $h = 4$ statistic is unavailable owing to insufficient observations in the auxiliary regression. Tests are interpreted at the nominal 5% level, without multiple-testing correction or adjustment for model selection.

At least one considered auxiliary order yields rejection of the null of no system residual serial correlation in each regime. This qualifies any conclusion that the selected models are fully dynamically adequate.

S5.2 Residual normality and autocorrelation plots

Figure S2 reproduces the thesis's residual Q–Q plots under the baseline $K = 4$ specification. The main article reports that Shapiro–Wilk normality is rejected in four of twelve equation/regime combinations, concentrated in the first two regimes. Tail departures are nevertheless visible in some equations with non-rejection.

Figure S3 preserves the residual ACF panels for the five equation/regime combinations rejected by the baseline Ljung–Box test at the nominal 5% level: UNRATE in regime 1; FEDFUNDS in regimes 2 and 3; UNRATE and FEDFUNDS in regime 4. Each panel shows first-order dependence; the regime-3 FEDFUNDS plot additionally shows negative dependence around lags 4 and 5.

These graphical diagnostics complement, but do not replace, the equation-wise test table in the main article. In particular, the selection and posterior-mean conditioning are not propagated through the conventional residual tests.

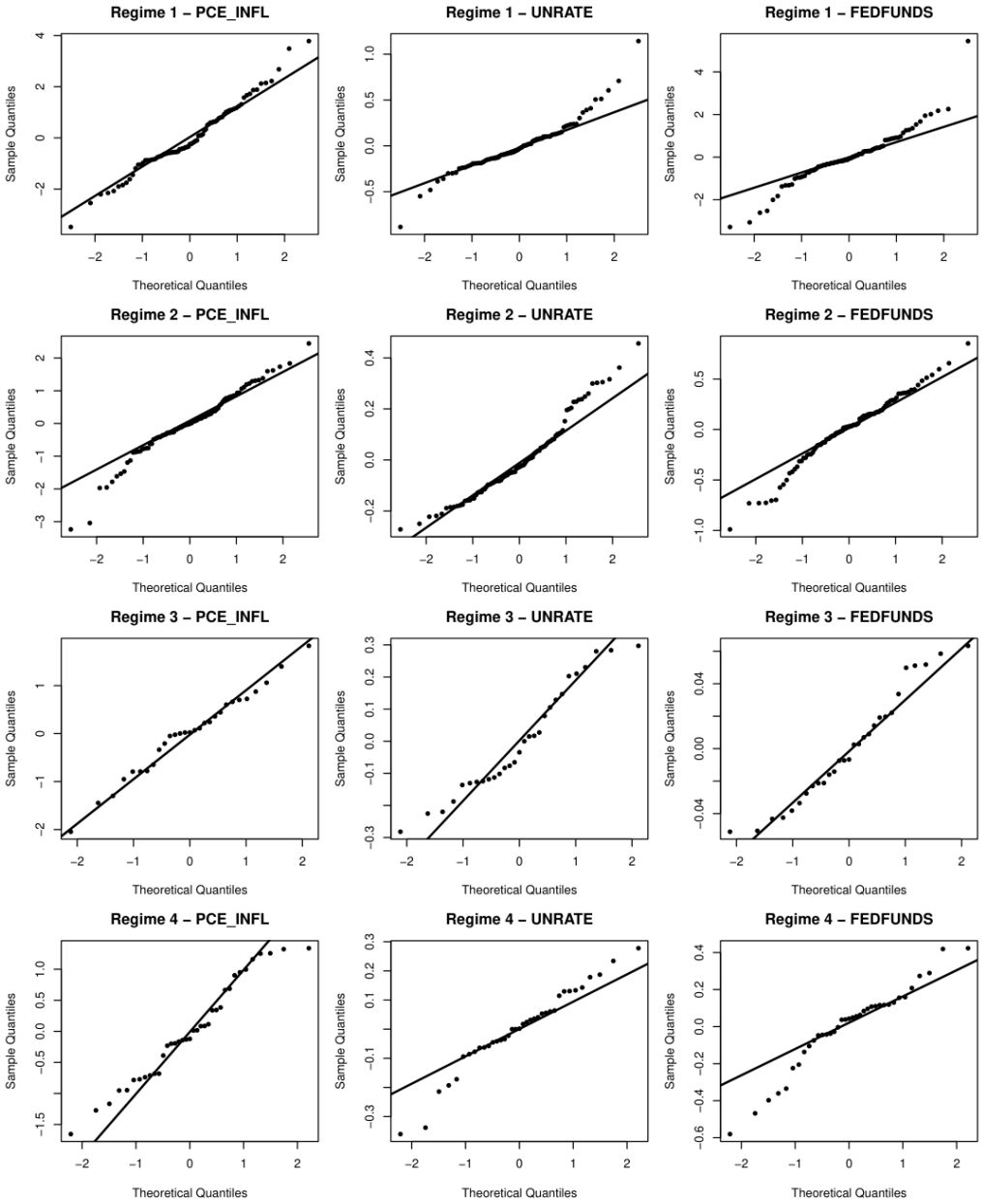


Figure S2: Residual Q–Q plots for the baseline $K = 4$ specification, copied from thesis Figure 6.4. These plots concern posterior-mean BVAR residuals conditional on each selected modal model.

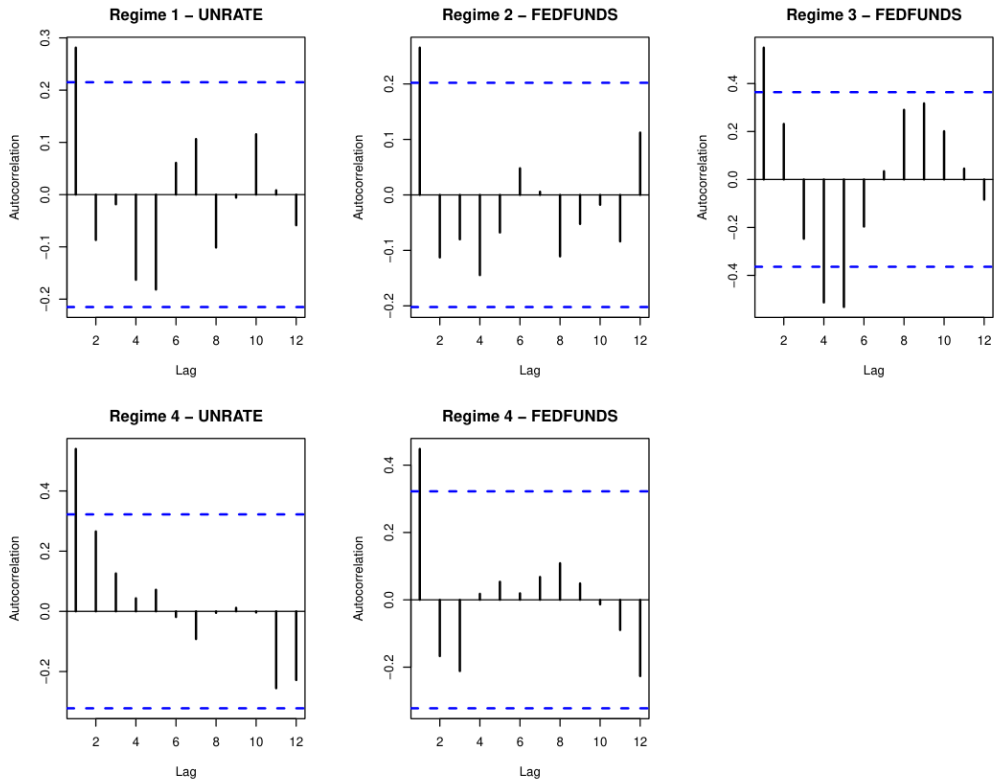


Figure S3: Residual autocorrelation functions for the five baseline $K = 4$ equation/regime combinations with a nominal Ljung–Box rejection. Extracted from thesis Figure 6.5; dashed lines are the original reference bounds.

S6 Additional Robustness Diagnostics, $K = 8$

The robustness specification increases the maximum candidate lag order to $K = 8$, giving $2^8 = 256$ blockwise lag configurations per regime. The selected break partition is unchanged, while all four modal lag configurations differ from those selected under $K = 4$. The fourth regime's posterior-mean modal-model companion spectral radius is 0.993, close to the unit-stability boundary. This scalar is not a stability probability over the posterior.

Figure S4 reproduces the regime-specific posterior inclusion probability plots from thesis Figure 6.6 for the eight potential lags. The figure shows why reporting a single modal binary sequence is an incomplete description of model uncertainty, especially as the number of candidate configurations rises.

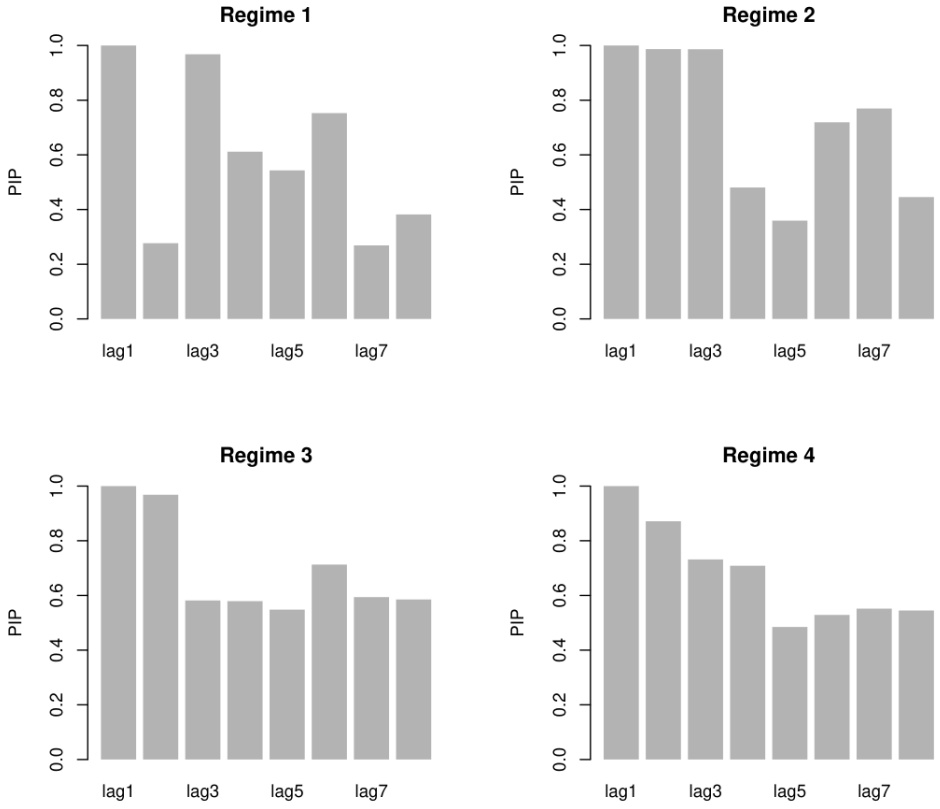


Figure S4: Posterior inclusion probabilities for lags 1–8 by selected empirical regime under $K = 8$, as in thesis Figure 6.6. They are conditional on the selected break partition and averaged over lag configurations.

Under $K = 8$, the total numbers of nominal 5% rejections are four (Shapiro–Wilk), three (Ljung–Box), and one (ARCH–LM), compared with four, five, and one in the baseline $K = 4$ analysis. Figure S5 shows the residual ACFs for the three remaining Ljung–Box rejections: UNRATE in regime 1 and FEDFUNDS in regimes 2 and 3. Residual first-order correlation remains visible and there are negative ACF values around lags 4 and 5 for FEDFUNDS in regime 3.

Since changing K also changes the common effective likelihood mask, the reduced number of rejections cannot be attributed solely to admitting more lags. The thesis does not report a directly comparable multivariate LM table for $K = 8$; none is created in this supplement.

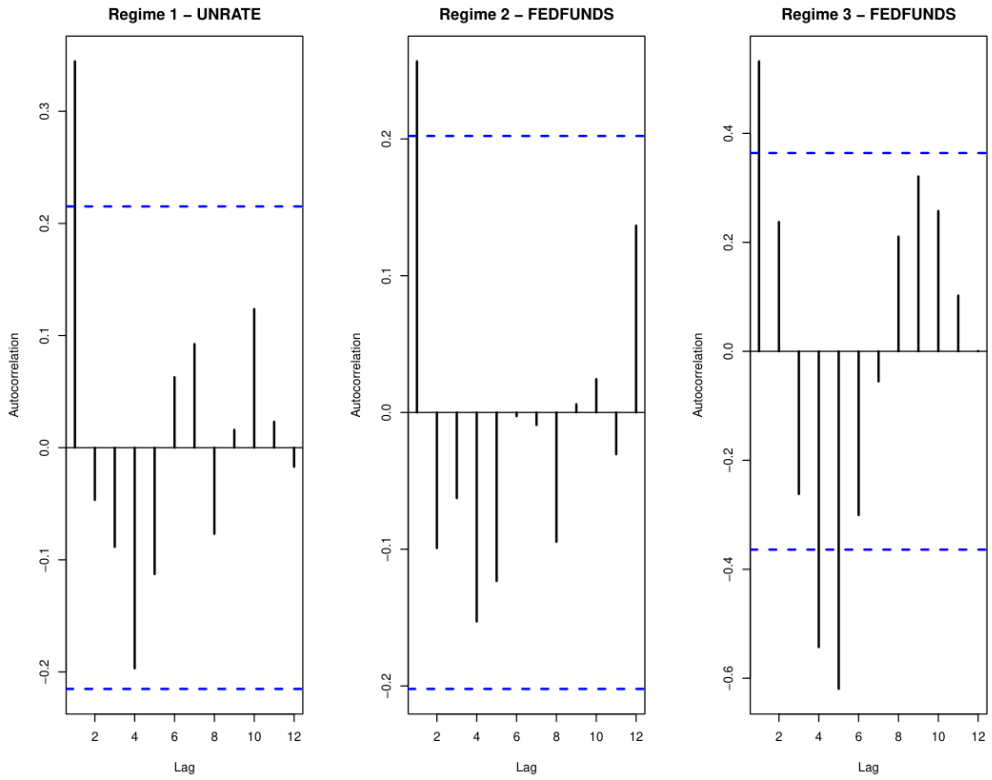


Figure S5: Residual ACFs for the three equation/regime combinations rejected by the Ljung–Box test under $K = 8$. Extracted from thesis Figure 6.7. The original reference bounds are unchanged.

S7 Additional Thesis Simulations Not Included in the Main Article

Status and scope. The main manuscript deliberately presents only the final three-break simulation. For completeness, Sections S7.1–S9 document the one-break and two-break experiments and the simulation-based comparison of alternative break counts reported in the original thesis, Chapter 5. They are *not* additional experiments newly run for the article. The original numerical tables and figures have been transcribed or extracted from thesis Sections 5.1.1–5.1.2 and 5.2.1–5.2.3, without claiming independent computational reproduction. These are distinct from the three-break study in the article and should not be pooled into a repeated-sampling performance assessment.

All these simulated systems are bivariate, use maximum lag order $K = 3$ (eight blockwise lag models per regime), a fixed training sample of $T_0 = 200$, and a minimum segment length of 200 observations. The shared hyperparameter values within each experiment are $\lambda_{\text{const}} = 10$, $\lambda_{\text{lag}} = 0.40$, $\lambda_{\text{decay}} = 1$, $\mu_{\text{own}} = 0$, and $\nu_0 = n + 2 = 4$. Each experiment calibrates its own fixed prior using its generated training sample, rather than sharing the training observations across different experiments. The base matrices and innovations of the first two regimes are as defined below. All break positions reported in these additional simulations are *estimation-sample indices*, unless specifically marked otherwise.

S7.1 One-break experiment: DGP and known break

The original R source initializes its random generator using seed 123 before the one-break simulation. The single-break DGP discards 200 initial simulation observations and retains $T = 4{,}200$ observations, comprising 200 training and 4,000 estimation observations. The true break is at $\tau_{\text{est}} = 2000$ (full-sample index 2200), splitting the estimation sample into two regimes of 2,000 observations each. Its true lag models are 101 and 010. The regime-specific intercepts are $c_1 = (0, 0)'$ and $c_2 = (0.20, -0.10)'$, while the nonzero autoregressive matrices are

$$\begin{aligned} B_{1,1} &= \begin{pmatrix} 0.45 & 0.10 \\ -0.05 & 0.35 \end{pmatrix}, \quad B_{3,1} = \begin{pmatrix} 0.18 & 0.00 \\ 0.08 & -0.15 \end{pmatrix}, \\ B_{2,2} &= \begin{pmatrix} -0.35 & 0.15 \\ 0.10 & 0.25 \end{pmatrix}, \quad B_{2,1} = B_{1,2} = B_{3,2} = 0_{2\times 2}. \end{aligned} \tag{10}$$

The innovation covariances are $\Sigma_1 = \left(\begin{smallmatrix} 1 & 0.30 \\ 0.30 & 1 \end{smallmatrix}\right)$ and $\Sigma_2 = \left(\begin{smallmatrix} 0.70 & -0.20 \\ -0.20 & 1.20 \end{smallmatrix}\right)$. The two companion-matrix spectral radii are 0.7680 and 0.6116, respectively (thesis Section 5.1.1).

With the break held at its true value, the first and second regime's true lag models receive posterior probabilities 0.9994 and 0.9949, respectively. Among the $8^2 = 64$ joint lag-model combinations, the true pair $(101, 010)$ receives probability 0.9943, compared with 0.0035 for the second-ranked pair $(101, 110)$. Figure S6 reproduces the five highest joint posterior probabilities.

S7.2 One-break experiment: unknown date on a local grid

The second exercise treats the break date as unknown but *restricts its support* to the 101 dates from 1950 to 2050, a preselected window of ± 50 observations around the known true location. The model-marginalized log evidence is maximized at $\tau_{\text{est}} = 2002$, with $\log m = -11{,}111.20$, compared with $-11{,}111.29$ at the true date 2000. Over this discrete local support, the posterior mode is 2002 (probability 0.1865), and the true date has probability 0.1715; the posterior mean and standard deviation are 2003.28 and 2.74, and the 95% equal-tailed credible interval is $[2000, 2010]$. The interval contains the true date. These posterior statements are *conditional on the restricted candidate grid*; they should not be described as a full-support analysis of the break location.

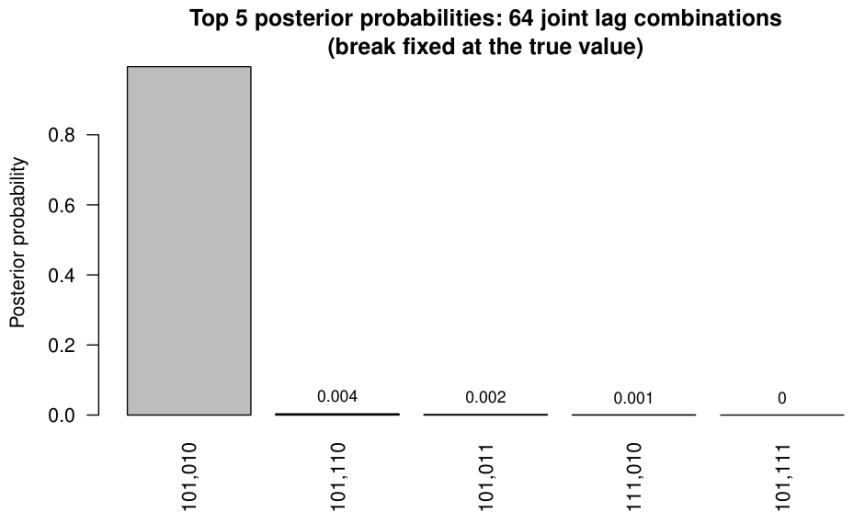


Figure S6: Five highest joint lag-configuration posterior probabilities with the one break fixed at its true date, as in thesis Figure 5.1. Probability is conditional on that known break.

S8 Two-Break Simulation: Exact Search and MCMC

S8.1 DGP and the two exact evaluations

The original R source initializes this experiment using seed 321. The two-break experiment discards 300 initial simulated observations and retains 2,600 observations: 200 for training and 2,400 for estimation. True estimation-sample break indices are $(800, 1600)$, or $(1000, 1800)$ including training. The three equal-length estimation regimes have true lag configurations $(101, 010, 110)$; the first two regimes retain the intercepts, active coefficient matrices, and innovation covariances of the one-break experiment in Section S7.1. For the new third regime,

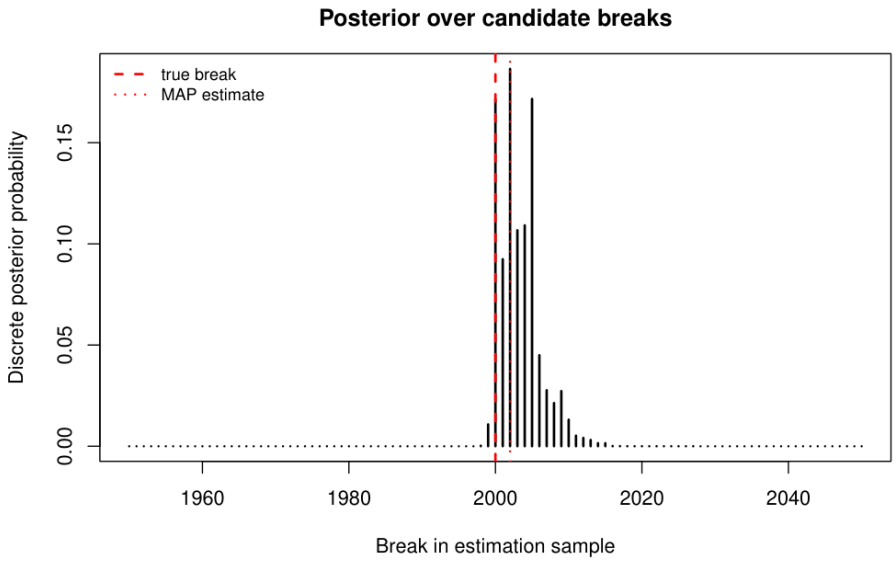
$$c_3 = \begin{pmatrix} -0.15 \\ 0.15 \end{pmatrix}, \qquad B_{1,3} = \begin{pmatrix} 0.25 & -0.10 \\ 0.05 & 0.30 \end{pmatrix}, \qquad B_{2,3} = \begin{pmatrix} -0.15 & 0.05 \\ 0.08 & -0.20 \end{pmatrix},$$
$$B_{3,3} = 0_{2\times 2}, \qquad \Sigma_3 = \begin{pmatrix} 1.10 & 0.15 \\ 0.15 & 0.80 \end{pmatrix}. \tag{11}$$

The three regime-specific spectral radii are 0.7680, 0.6116, and 0.5061. There are $8^3 = 512$ joint lag configurations (thesis Section 5.1.2).

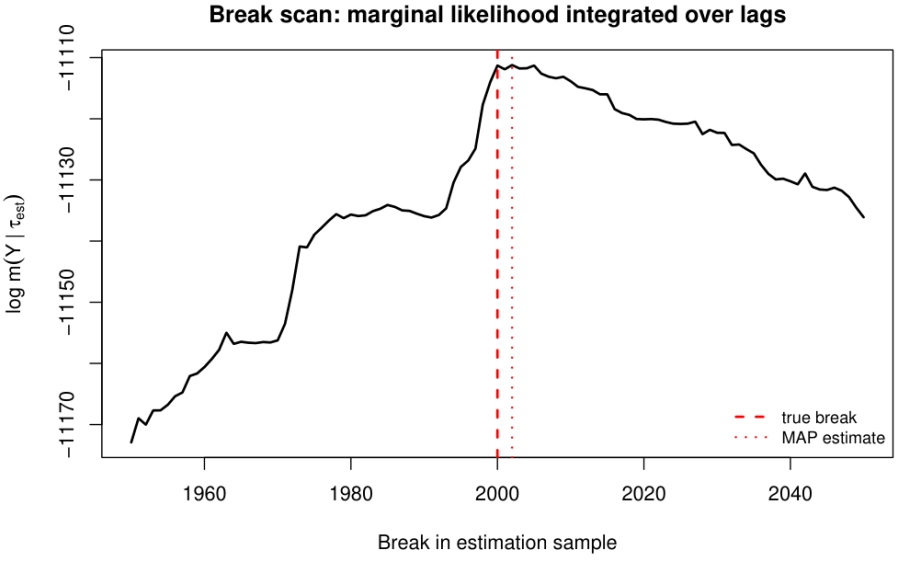
At the true break dates, joint lag model $(101, 010, 110)$ has probability 0.7966; runner-up $(101, 010, 111)$ has 0.1951. The next exercise evaluates the break pair *only* on a local Cartesian grid: 31 integer candidate dates (785–815) for the first break and 31 (1585–1615) for the second, giving $31^2 = 961$ pairs. The highest model-marginalized log evidence is at $(796, 1599)$, close to the true $(800, 1600)$. The two marginal posterior means are 800.05 and 1599.36 with standard deviations 3.64 and 3.36; the marginal modes are 796 and 1599. Conditional on the maximizing pair, the true joint lag configuration has posterior probability 0.8011 and the runner-up $(101, 010, 111)$ has 0.1908.

S8.2 Shared- and dispersed-start MCMC designs

The third exercise validates a collapsed MCMC procedure by comparison with the exact posterior computed on *the same two-break local support*. Four chains are run for 20,000 iterations, with 4,000 burn-in iterations; thinning by four applies to plotting/reduced representations. The

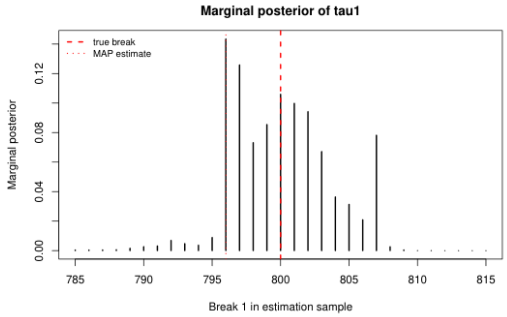


(a) Posterior distribution over the candidate break locations.

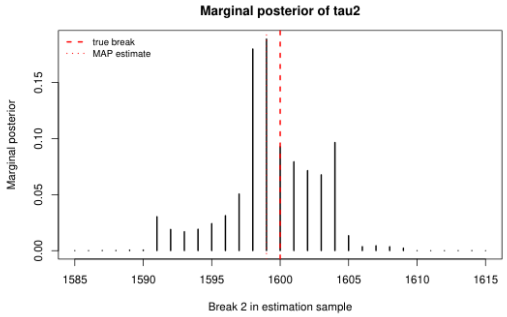


(b) Model-marginalized log marginal likelihood over the candidate break locations.

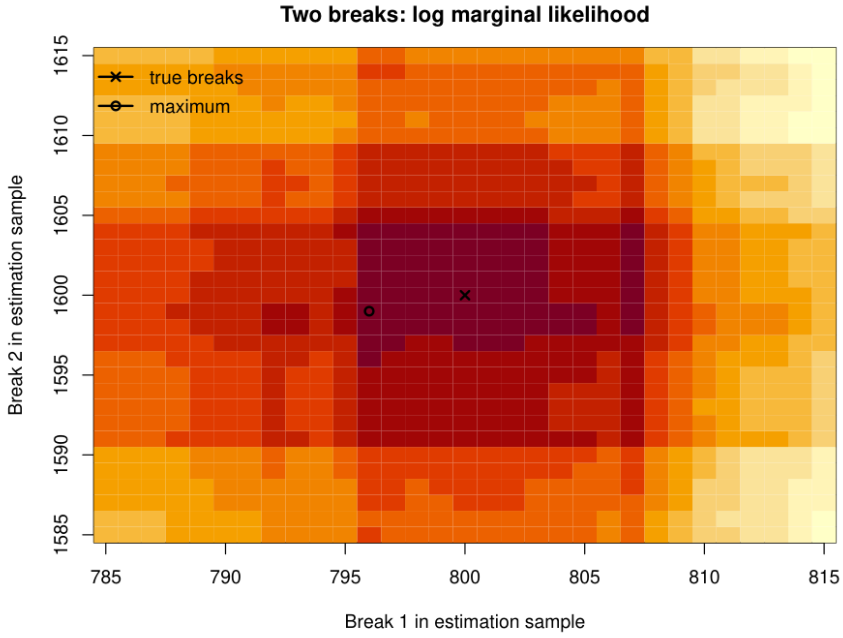
Figure S7: Single-break local-grid posterior probabilities (top) and model-marginalized log evidence (bottom), as in thesis Figure 5.2. The support is restricted to 101 candidate dates centered on the true break.



(a) Marginal posterior distribution of the first break.



(b) Marginal posterior distribution of the second break.



(c) Model-marginalized log marginal likelihood over the two-break grid.

Figure S8: Exact local-grid two-break comparison: the two marginal break-date posteriors and the heat map of model-marginalized log evidence, reproduced from thesis Figure 5.3. "Exact" refers to enumeration on the *restricted* 31×31 support, not the full set of admissible partitions.

shared-start implementation initializes each chain at estimation-sample break pair $(795, 1605)$ with lag models $(111, 111, 111)$ and seeds $(11, 22, 33, 44)$. The dispersed-start implementation uses the following break pairs and lag models, with seeds $(111, 222, 333, 444)$ in chain order:

$$\begin{aligned} &(788, 1588) : (111, 111, 111), \qquad (812, 1612) : (000, 000, 000), \\ &(792, 1608) : (101, 010, 110), \qquad (808, 1592) : (011, 001, 111). \end{aligned} \tag{12}$$

The original R implementation stores break positions after adding the $T_0 = 200$ training offset, as in the three-break exercise (Section S2). All starting configurations above use the thesis's estimation-sample coordinates. The support restriction to the scan's candidate grid is part of the experimental design and is also imposed on these MCMC chains.

Table S6: Break-date recovery and MCMC convergence for the two-break experiment (thesis Table 5.1).

Start	Break	True	Post. mean	Post. mode	95% interval	Mass within ±5	$\widehat{R}$	Bulk-ESS	Tail-ESS
Shared	τ_1	800	799.9862	796	[794, 807]	0.8763	1.0025	2207.77	2292.49
Shared	τ_2	1600	1599.4246	1598	[1591, 1605]	0.8906	1.0029	2149.76	2640.00
Dispersed	τ_1	800	800.0070	796	[794, 807]	0.8694	1.0014	2363.74	2704.20
Dispersed	τ_2	1600	1599.3450	1598	[1591, 1605]	0.8881	1.0023	2307.74	2558.91

Table S7: Marginal lag-model recovery in each regime, under the two MCMC starting designs (thesis Table 5.2).

Initialization	Regime	True γ_s	Modal γ_s	$P(\gamma_s = \gamma_s^0 \mid Y)$
Shared	1	101	101	0.9976
Shared	2	010	010	0.9951
Shared	3	110	110	0.8012
Dispersed	1	101	101	0.9980
Dispersed	2	010	010	0.9951
Dispersed	3	110	110	0.7974

Table S8: Individual lag-indicator PIPs and MCMC diagnostics in the two-break experiment (thesis Table 5.3).

Initialization	Regime	PIP (lags 1–3)	$\widehat{R}$	Bulk-ESS	Tail-ESS	$T_{\min}$
Shared	1	(1.000, 0.002, 1.000)	(NA, 1.002, NA)	(NA, 1725.3, NA)	(NA, 1725.3, NA)	(0, 0, 0)
Shared	2	(0.000, 1.000, 0.005)	(1.001, NA, 1.003)	(5121.7, NA, 1325.6)	(5121.7, NA, 1325.6)	(0, 0, 2)
Shared	3	(1.000, 1.000, 0.199)	(NA, NA, 1.001)	(NA, NA, 2413.9)	(NA, NA, NA)	(0, 0, 326)
Dispersed	1	(1.000, 0.002, 1.000)	(NA, 1.002, NA)	(NA, 2261.5, NA)	(NA, 2261.5, NA)	(0, 0, 0)
Dispersed	2	(0.000, 1.000, 0.005)	(NA, NA, 1.002)	(NA, NA, 1975.7)	(NA, NA, 1975.7)	(0, 0, 0)
Dispersed	3	(1.000, 1.000, 0.203)	(NA, NA, 1.002)	(NA, NA, 2547.2)	(NA, NA, NA)	(0, 0, 338)

Notes. Undefined $\widehat{R}$ and ESS are retained as "NA", as in the thesis. $T_{\min}$ is the smallest number of inclusion-state switches across the four chains. In particular, the third lag of regime 3 has PIP 0.199 or 0.203, with at least 326 or 338 switches. Undefined diagnostics or zero transitions for almost-constant indicators do not themselves establish adequate mixing.

S8.3 Acceptance rates and comparison with the exact posterior

The following comparison concerns posterior probabilities on *exactly the same restricted grid*; thus it measures MCMC approximation accuracy against the discrete local target, not against a posterior over all admissible break positions. Total variation (TV) is one half the L_1 distance. The numerical values and rounding follow thesis Table 5.5.

Table S9: Mean proposal acceptance rates over four chains, two-break MCMC (thesis Table 5.4).

Initialization	Overall	Break-date proposals	Lag-model proposals
Shared	0.2658	0.4849	0.0468
Dispersed	0.2656	0.4848	0.0464

Table S10: Distance between exact local-grid and MCMC posterior probabilities for two breaks (thesis Table 5.5).

Start	Distribution	L_1	TV	Max. absolute error	RMSE	Correlation
Shared	τ_1 marginal	0.042964	0.021482	0.008748	0.002454	0.998458
Shared	τ_2 marginal	0.050125	0.025062	0.011054	0.002985	0.998326
Shared	(τ_1, τ_2) joint	0.125749	0.062875	0.007567	0.000414	0.990895
Shared	γ_1	0.001243	0.000621	0.000621	0.000310	1.000000
Shared	γ_2	0.003027	0.001513	0.001513	0.000685	0.999999
Shared	γ_3	0.013249	0.006625	0.006625	0.003312	0.999948
Shared	$(\gamma_1, \gamma_2, \gamma_3)$ joint	0.015304	0.007652	0.007019	0.000414	0.999949
Dispersed	τ_1 marginal	0.037968	0.018984	0.011776	0.002563	0.998350
Dispersed	τ_2 marginal	0.058717	0.029359	0.012710	0.003396	0.998147
Dispersed	(τ_1, τ_2) joint	0.123458	0.061729	0.004034	0.000367	0.992884
Dispersed	γ_1	0.000430	0.000215	0.000215	0.000107	1.000000
Dispersed	γ_2	0.003152	0.001576	0.001576	0.000688	0.999999
Dispersed	γ_3	0.020874	0.010437	0.010437	0.005219	0.999869
Dispersed	$(\gamma_1, \gamma_2, \gamma_3)$ joint	0.023054	0.011527	0.011300	0.000667	0.999867

The exact and dispersed-start MCMC marginal break-date posteriors overlap in thesis Figure 5.4, reproduced here as Figure S9. The maximum joint break-date TV is about 0.063 for both starting designs, while the smallest reported correlation in the joint break-date comparisons is 0.990895 (shared start). The smaller lag-model distances in Table S10 are evidence of a close approximation to this particular local posterior target, not a general guarantee of MCMC performance.

S9 Selecting the Break Count in the Two-Break Simulation

The final supplementary experiment uses the same two-break DGP from Section S8, but treats the number of breaks as unknown among $m = 0, 1, 2$. Unlike the exact local grid centered on the true pair in Section S8, this exercise uses an *ex ante coarse-to-fine search* that does not center the search grid at the known DGP breaks. For $m = 1$ and $m = 2$, the algorithm uses minimum segment length 200, coarse grid step 25 and refinement radius 30 (thesis Sections 5.1.2 and 5.2.3). Equal prior probabilities for the three candidate counts are stated in the simulation design; the statistic tabulated here is nonetheless a *profile* log marginal likelihood, maximized over the searched break partitions, and must not be represented as a posterior probability of a break count.

The largest profile score is attained at the correct count $m = 2$, selecting $(796, 1599)$ rather than the true $(800, 1600)$. The reported three regime-specific modal lag models match the DGP. These findings refer to this *single simulated dataset* and the stated coarse-to-fine search; they are not estimates of a repeated-sampling break-count selection rate. The results also must not be conflated with the exhaustive or dynamic-programming search used in the main paper's empirical application.

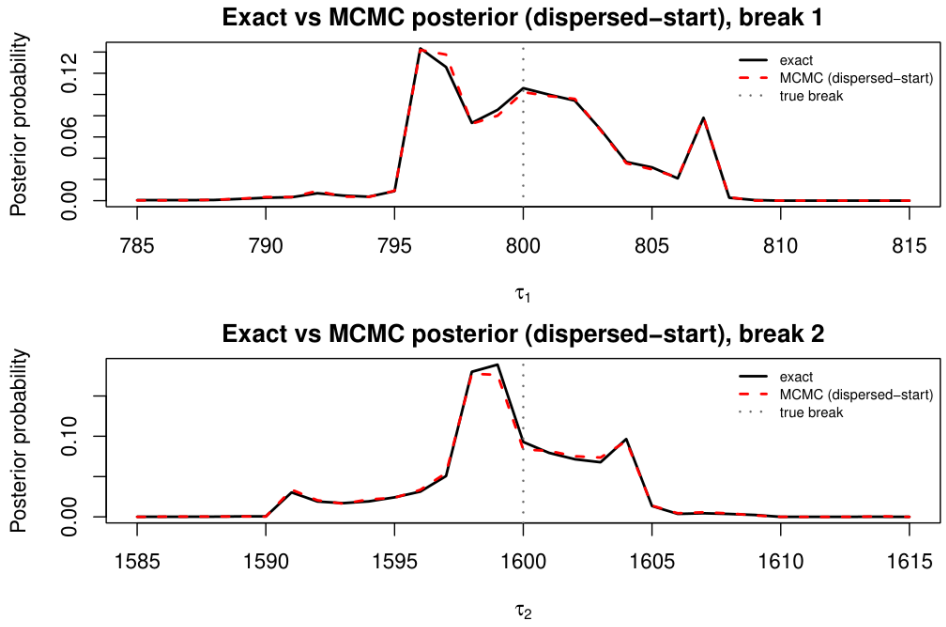


Figure S9: Exact versus dispersed-start MCMC marginal break-date posterior distributions on the identical local two-break support, reproduced from thesis Figure 5.4.

Table S11: Profile marginal-likelihood comparison of break counts for the two-break DGP (thesis Table 5.6).

m	Profile log ML	Δ from best	Selected break(s)	Modal lag models by regime
0	−6959.975	−237.1074	—	111
1	−6864.574	−141.7067	790	101, 110
2	−6722.867	0.0000	796, 1599	101, 010, 110

S10 Code, Input Snapshots, and Provenance

The R code and frozen inputs for the main article and this Supplementary Material are hosted in the article repository on GitHub. The current repository provides `code/paper/` for the selected three-break simulation and empirical application, `code/supplementary/simulation_experiments.R` for the historical one-break, two-break, and break-count experiments of Sections S7–S9, and the three original FRED CSV snapshots in `data/raw/`. The respective execution wrappers are `scripts/run_paper.ps1` and `scripts/run_supplementary.ps1`. The R dependencies are recorded in `renv.lock`; restoration instructions and verification limitations appear in `docs/ENVIRONMENT.md`. The original full thesis script remains in the separate historical dissertation project and is not mirrored in the current article repository. Its README and Git history document source provenance. In particular, the supplementary simulation file retains the original scientific R statements without modification.

The numerical summaries and existing figures in this document were checked against the original thesis, specifically Sections 5.1.1–5.1.3, 5.2.1–5.2.4 and 6.1–6.4, Tables 5.1–5.10 and 6.2/6.7, and Figures 5.1–5.6 and 6.4–6.7. The thesis reports the three-break MCMC starting dates relative to the estimation sample, whereas the R implementation includes the training sample; their difference is exactly $T_0 = 200$, as shown in Table S1. Figures were cropped from the thesis without changing plotted content.

As documented in the repository README and a historical C09 verification record retained in Git history, an author-executed numerical comparison of *selected main-article outputs* agreed with the thesis at reported table precision on 15 October 2026 (Windows 11, R 4.6.1, `posterior` 1.7.0). This is not an independent third-party replication or evidence of bitwise equivalence. The supplementary execution route has been provided, but a fresh execution and comparison covering *all* tables and figures in Sections S7–S9 have not been established. Those sections report existing thesis experiments, not additional simulations carried out for this Supplementary Material. No new data revisions, statistical tests, or numerical results were introduced in preparing this document.